

Objective

To build a Python-based CSV Data Processor that can read CSV files, analyze numerical data, calculate useful statistics, and handle missing or invalid values.

Learning Objectives

- Understand CSV file processing in Python.
- Learn to use Python's built-in csv module.
- Practice reading structured data.
- Perform calculations on numerical data.
- Handle missing values.
- Handle invalid numerical values.
- Practice loops and conditional statements.
- Build an interactive interface using Streamlit.
- Generate a simple summary report.

3. Key Features

CSV File Upload

Users can upload .csv files directly through the Streamlit interface.

Dataset Preview

The uploaded records and columns are displayed in the application.

Numeric Column Detection

The application automatically identifies columns containing numerical values.

Statistical Analysis

For the selected column, the application calculates:

- Total number of records
- Valid values
- Total
- Average

- Highest value
- Lowest value
- Missing values
- Invalid values

Error Handling

The application handles:

- Empty CSV files
- Missing values
- Invalid numeric values
- Invalid file encoding
- CSV files without columns

Summary Report

A readable summary report is generated containing the calculated statistics.

Download

Users can download the generated summary report as a .txt file.

Technologies Used

Technology	Purpose
Python	Core programming
Streamlit	Web interface
CSV module	Reading and processing CSV files
Python Lists	Storing records and numeric values
Loops	Processing CSV records

Exception Handling Handling invalid input/errors

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY20>

```
import streamlit as st
import csv
import io

st.set_page_config(
    page_title="CSV Data Processor",
    page_icon="📊",
    layout="wide"
)

st.title("📊 CSV Data Processor")
st.write(
    "Upload a CSV file to analyze records and generate "
    "useful statistical summaries."
)

st.divider()
uploaded_file = st.file_uploader(
    "📁 Upload a CSV file",
    type=["csv"]
)

if uploaded_file is not None:
    try:
        # Read uploaded CSV
        content = uploaded_file.getvalue().decode("utf-8")

        reader = csv.DictReader(io.StringIO(content))
        rows = list(reader)

        if not rows:
            st.warning("The CSV file is empty.")
            st.stop()

        columns = reader.fieldnames

        if not columns:
            st.error("No columns found in the CSV file.")
            st.stop()
        st.subheader("📋 Dataset Information")

        col1, col2, col3 = st.columns(3)

        with col1:
            st.metric("Total Records", len(rows))

        with col2:
            st.metric("Total Columns", len(columns))

    except Exception as e:
        st.error(f"An error occurred: {e}")

with col3:
    st.metric("File Name", uploaded_file.name)
    st.subheader("📈 Uploaded Data")

st.dataframe(
    rows,
    use_container_width=True
)

st.subheader("📊 Statistical Analysis")

numeric_columns = []

for column in columns:
    valid_values = []

    for row in rows:
        value = row.get(column, "").strip()

        if value:
            try:
                valid_values.append(float(value))
            except ValueError:
                pass

    if valid_values:
        numeric_columns.append(column)

if not numeric_columns:
    st.warning(
        "No numeric columns were found in the uploaded CSV."
    )
    st.stop()

selected_column = st.selectbox(
    "Select a numeric column to analyze:",
    numeric_columns
)

valid_values = []
invalid_count = 0
missing_count = 0

for row in rows:
    value = row.get(selected_column, "").strip()

    if value == "":
        missing_count += 1
```

Output: <http://localhost:8501>



CSV Data Processor

Upload a CSV file to analyze records and generate useful statistical summaries.

📁 Upload a CSV file



Upload

200MB per file • CSV



Upload a CSV file to start processing.



Example CSV Format

```
Employee,Department,Salary
John,IT,45000
Priya,HR,40000
Rahul,Finance,50000
Anita,IT,55000
```

The processor can analyze numeric columns such as Salary, Sales, Marks, Quantity, Revenue, etc.

Upload a CSV file to analyze records and generate useful statistical summaries.

 10.1002/9781118471996.ch10

Total Records:

Up

Index	Customer id	
-------	-------------	--

Select a numeric column to analyze:

Tricks

1000

500

500

1.00

File Name : customers

```
Total Records : 10000
Total Columns : 12
```

11.2.2.2. *Phylogenetic analysis*

Invalid Values : 0

Highest : 1,000.00

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Age Group	Very bad	Bad
18-29	10%	10%
30-49	10%	10%
50-69	10%	10%
70+	10%	10%

Linkedin:- <https://lnkd.in/p/dcuqG7Df>

Statistical Analysis

Select a numeric column to analyze

Index

Valid Count

1000

Total

500,500.00

Average

500.50

Highest

1,000.00

Lowest Value: 0.00

Missing Values: 0

Invalid Values: 0

Summary Report

CSV Data Processing Summary

File Name : customers-1000.csv

Total Records : 1000

Total Columns : 12

Analyzed Column : Index

Valid Values : 1000

Missing Values : 0

Invalid Values : 0

Total : 500,500.00

Average : 500.50

Highest : 1,000.00

Lowest : 0.00

Download Summary Report

Shambhavi E L • You

Student at The Oxford College of ...

Just now

VEDA Internship – Day 20 Completed! Successfully completed Day 20 of my VEDA Internship – Python Programming Track. Todays task was Build a CSV...

more

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